

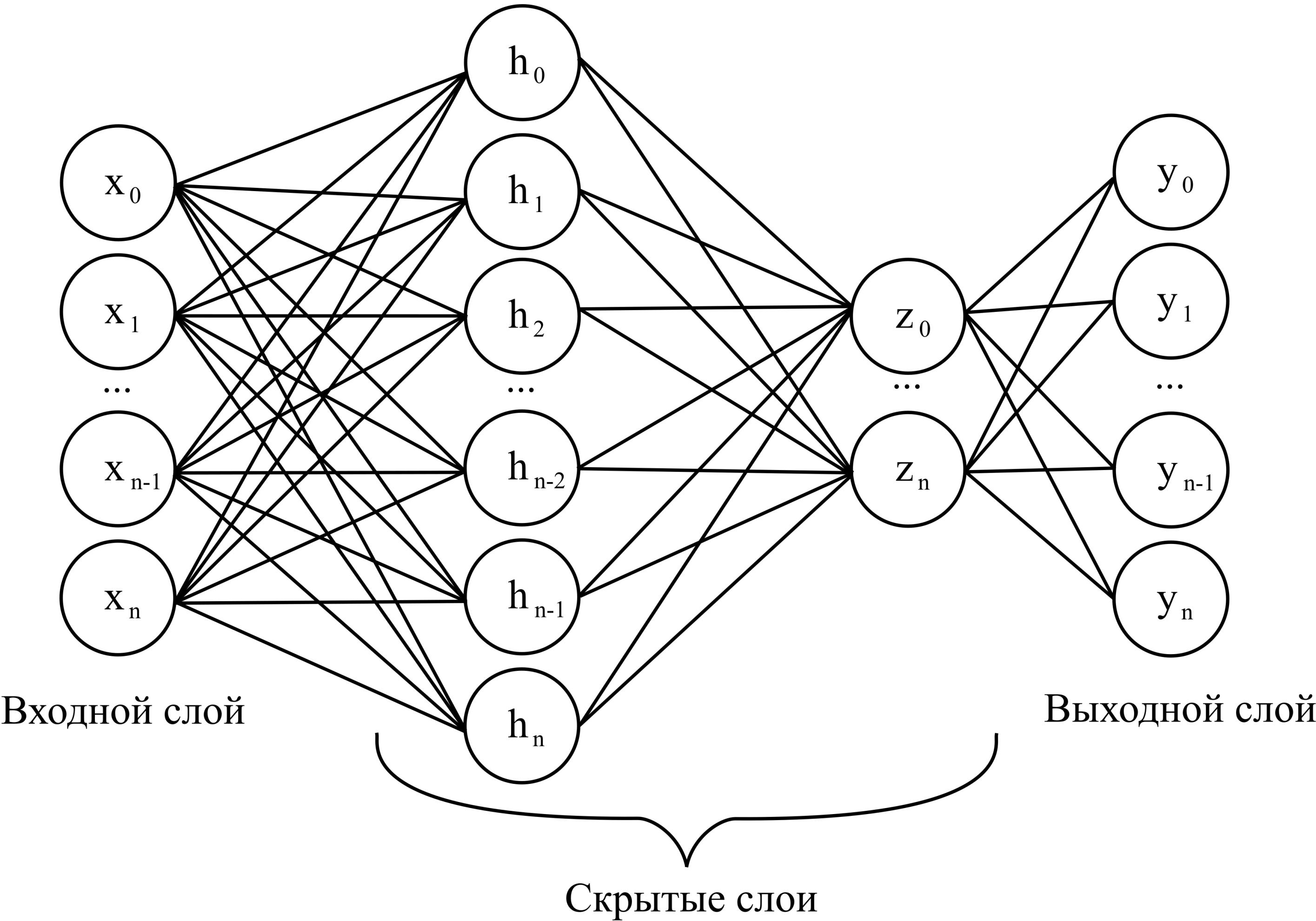
IP-ядро нейронной сети прямого распространения для распознавания рукописных цифр

Постановка задачи

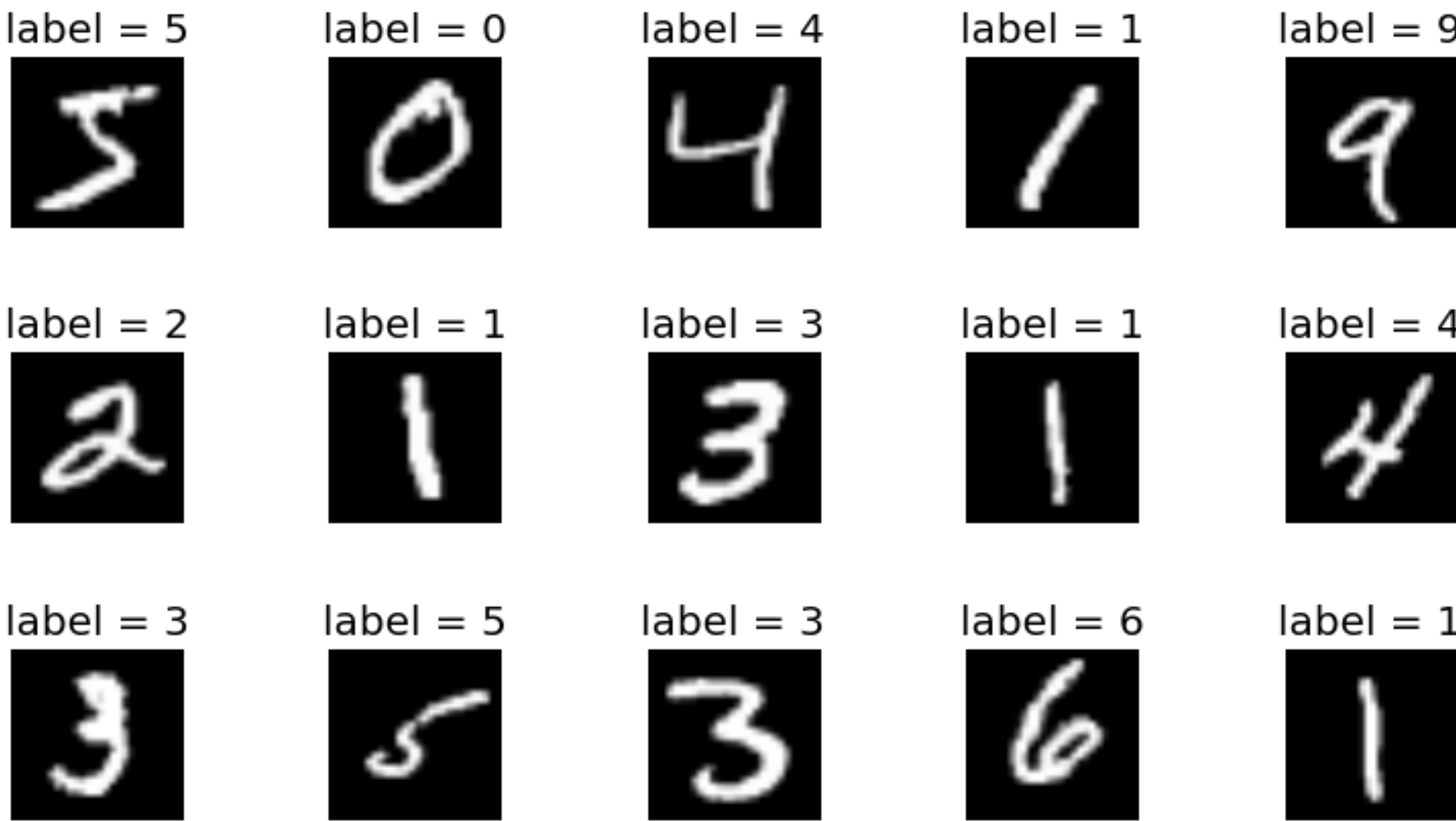
Таблица сравнения точности и параметров аналогичных разработок

Автор	Архитектура НС	# Число параметров	Точность
Liang, et al. (2018)	784-2048-2048-2048-10	10 100 000	98.32%
Umuroglu, et al. (2017)	784-1024-1024-10	1 863 690	98.40%
Medus, et al. (2019)	784-600-600-10	891 610	98.63%
Huynh	784-126-126-10	115 920	98.16%
Huynh	784-40-40-40-10	34 960	97.20%
Westby, et al.	784-12-10	9 550	93.25%

Пример полносвязной архитектуры



Пример изображений



Сверточная нейронная сеть, точность 88% и 23520 параметров

		Detected class									
		0	1	2	3	4	5	6	7	8	9
Target Class	0	99.5%	0.0%	0.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.2%
	1	3.6%	89.4%	0.5%	1.0%	2.0%	0.0%	0.0%	3.1%	0.2%	0.2%
	2	4.1%	1.6%	90.0%	2.0%	0.3%	0.2%	0.0%	0.4%	0.8%	0.6%
	3	1.1%	0.3%	2.4%	90.8%	0.1%	0.2%	0.1%	0.4%	0.3%	4.3%
	4	1.3%	0.3%	0.0%	0.1%	97.6%	0.0%	0.4%	0.2%	0.1%	0.0%
	5	1.2%	0.0%	0.3%	22.0%	0.4%	68.0%	2.4%	0.2%	0.7%	4.8%
	6	6.4%	0.4%	0.4%	1.1%	6.2%	0.6%	82.4%	0.7%	1.0%	0.8%
	7	0.5%	0.4%	7.4%	0.2%	0.4%	0.0%	0.5%	90.1%	0.0%	0.5%
	8	1.5%	0.1%	0.2%	6.7%	1.8%	0.5%	0.9%	0.9%	87.3%	0.1%
	9	8.7%	0.2%	0.0%	0.8%	0.7%	0.3%	0.0%	3.9%	0.5%	84.9%

Giardino D. et al. FPGA implementation of hand-written number recognition based on CNN // International Journal on Advanced Science, Engineering and Information Technology. – 2019. – Т. 9. – No. 1. – P. 167-171.